



!"!#\$!%



Tópicos

SSH (Secure shell)

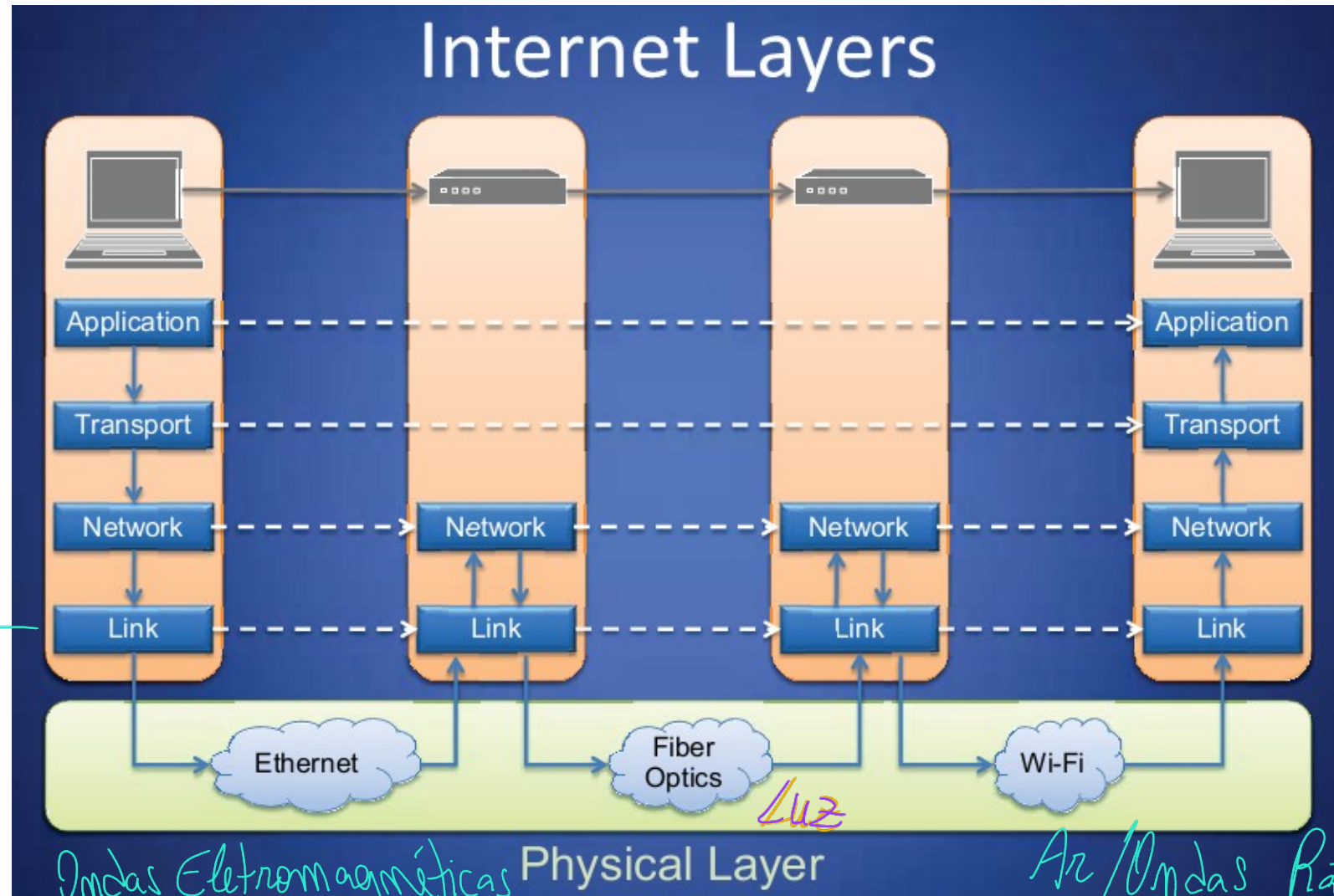
SSL/TLS (Secure Socket Layer / Transport Layer Security)

HTTPS (HTTP over SSL/TLS)

Virtual Private Networks (VPN)



Pilha protocolar TCP/IP





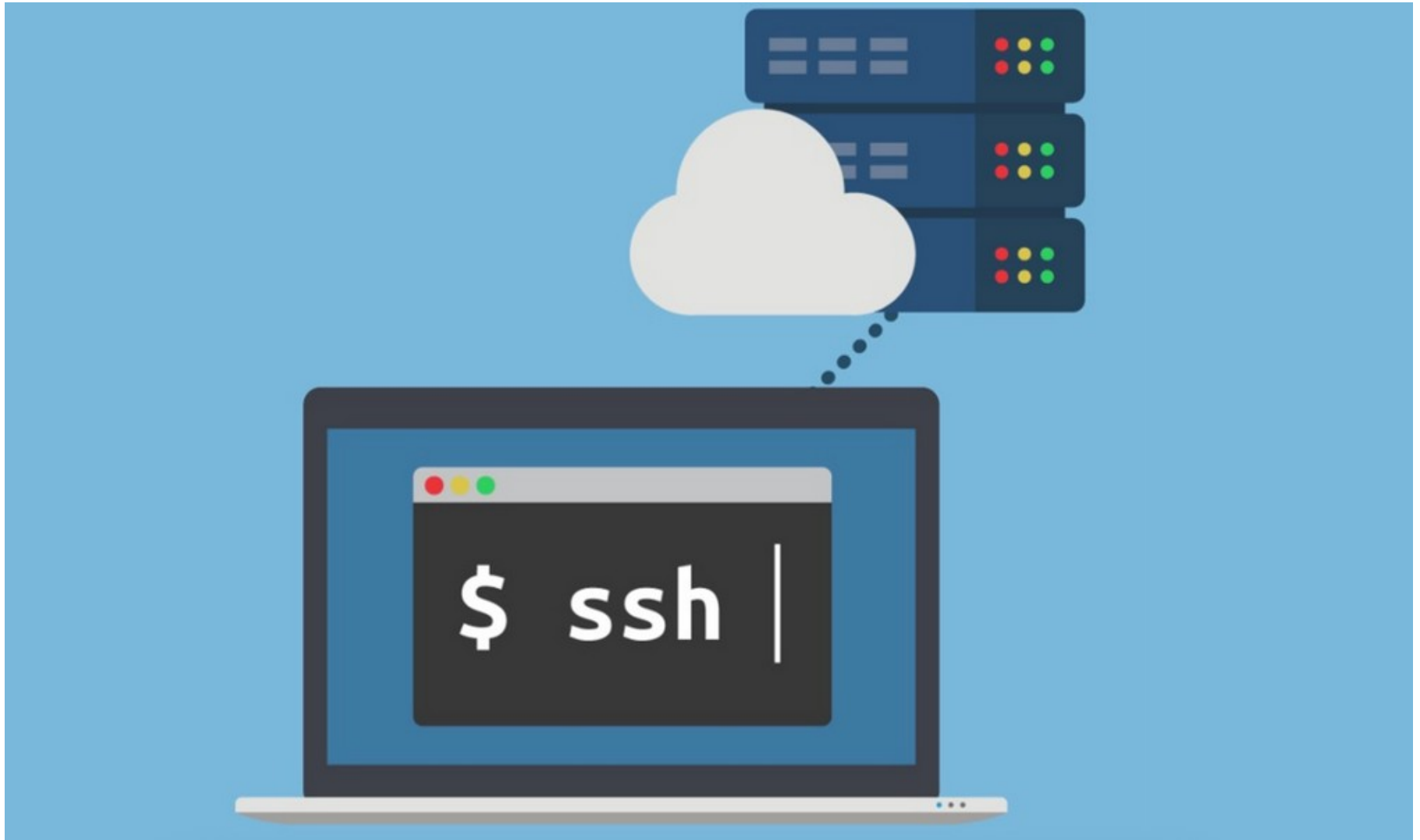
Protocolos de segurança:

Localização na pilha protocolar TCP/IP

Pilha TCP/IP	Aproximação	Protocolos
Aplicação	Cifra entre extremos (end-to-end encryption)	HTTPS, SSH, Kerberos
Transporte		SSL/TLS
Internet		IPSec
Acesso à Rede	Cifra por troços (link encryption)	Ligações sem fios, PPP <i>Point-to-point protocol</i>



SSH (Secure Shell)



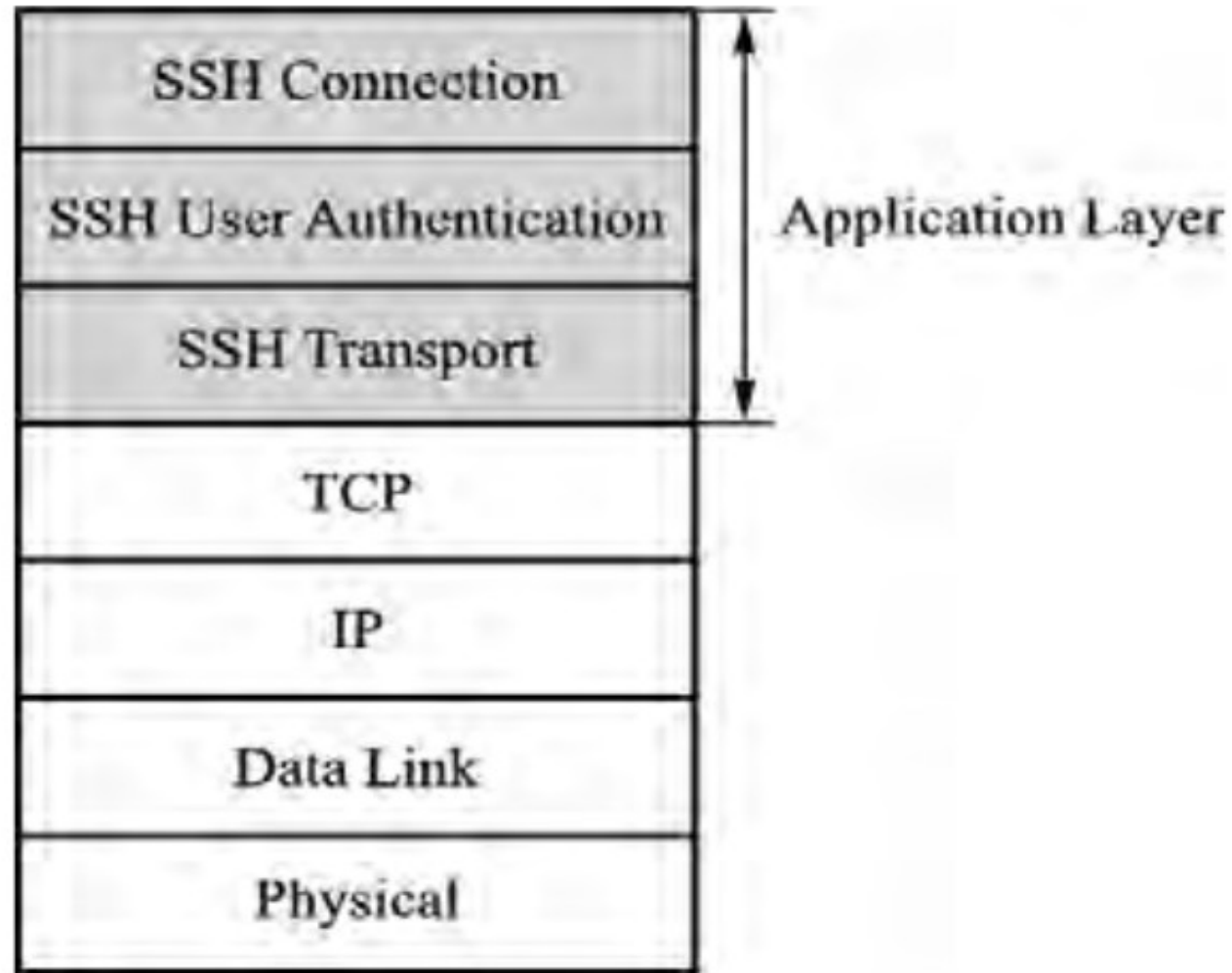


!

" # \$

%

" & "





&

(" " "

) #

* + # "

, # # -

& . %

%) " "- /

"- -

&

0 1 # +

2



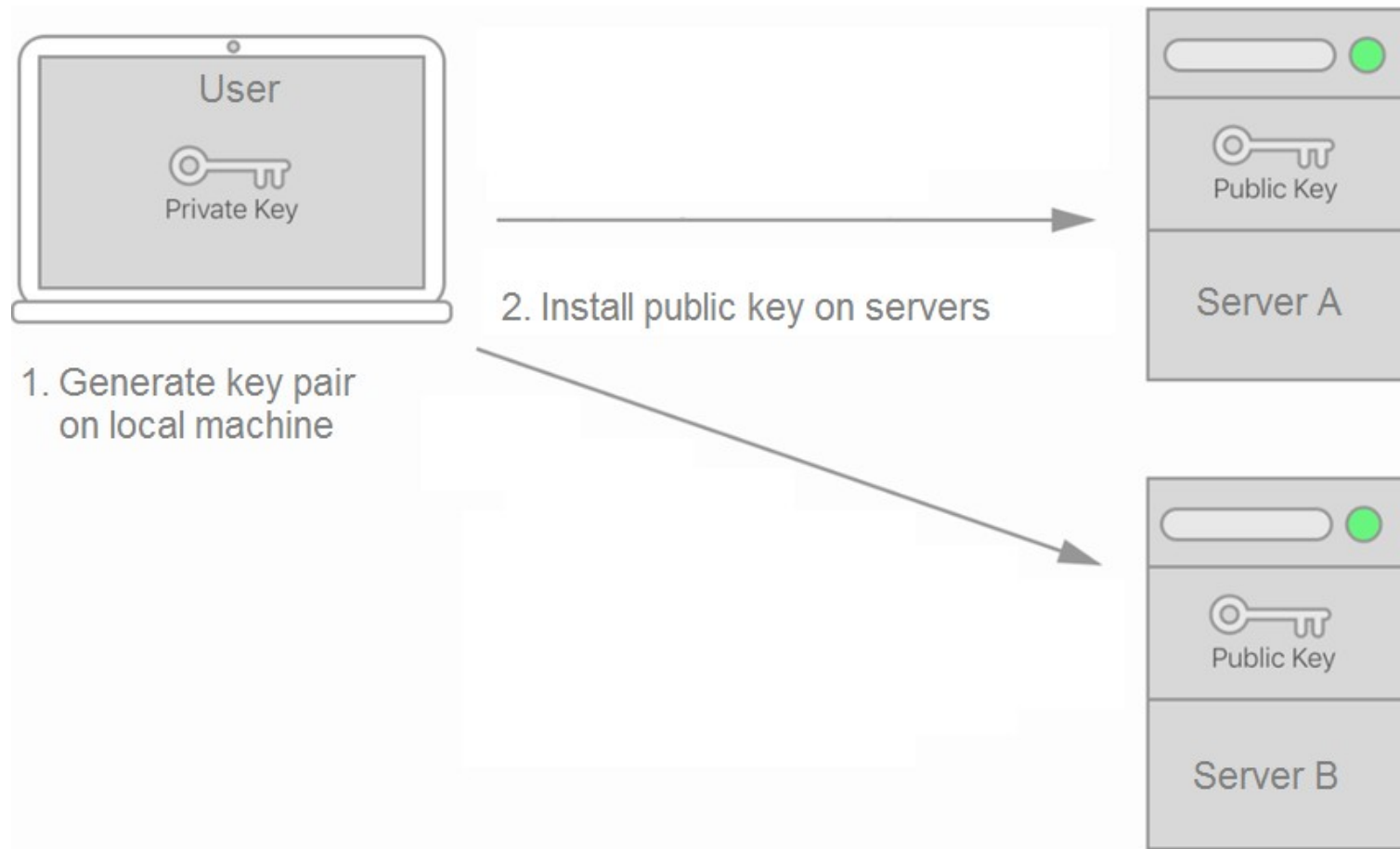
% "-

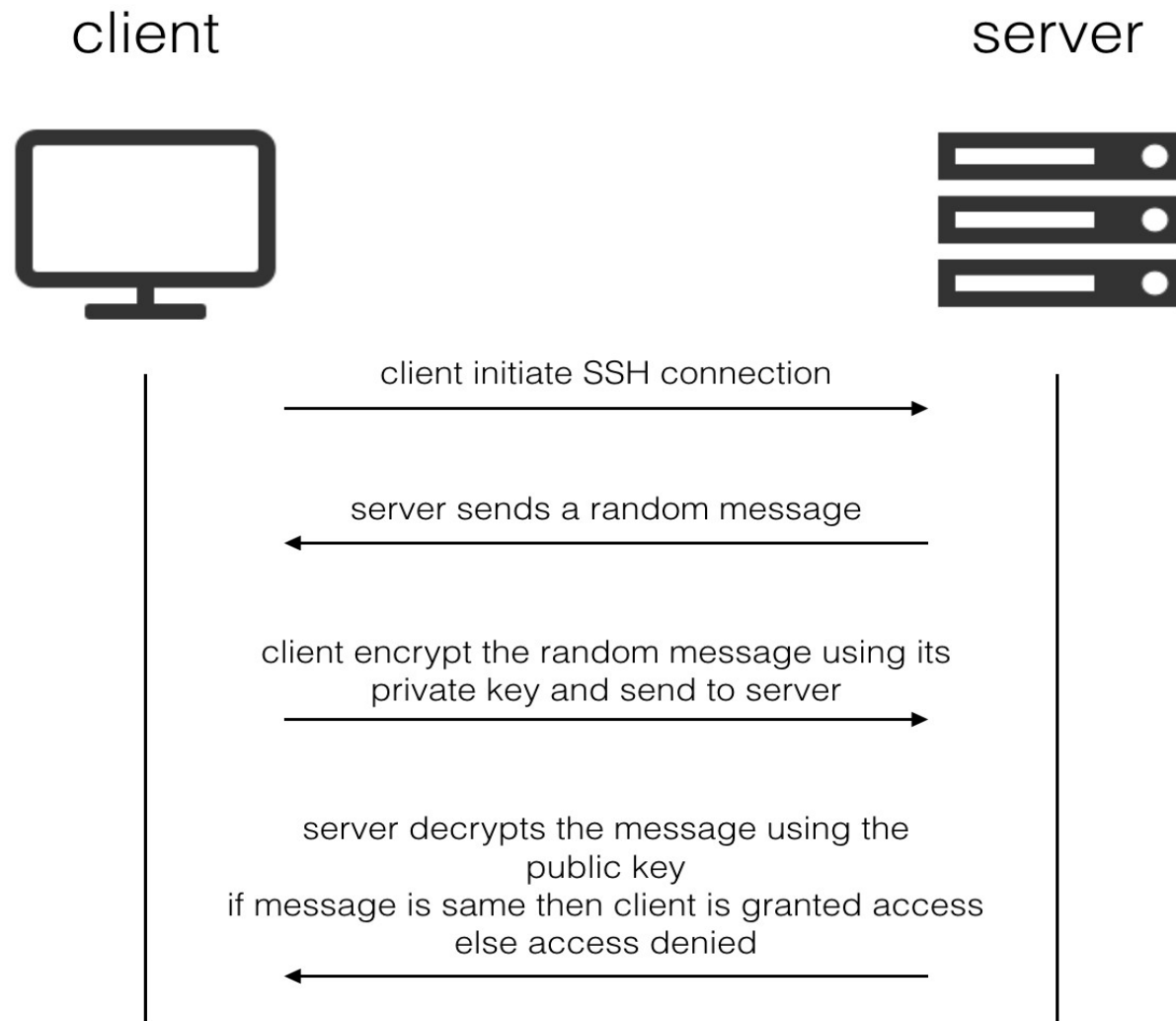
```
The authenticity of host '111.111.11.111 (111.111.11.111)' can't be established.  
ECDSA key fingerprint is fd:fd:d4:f9:77:fe:73:84:e1:55:00:ad:d6:6d:22:fe.  
Are you sure you want to continue connecting (yes/no)? yes
```

, " 3 " 1
") 2
4))
" 5
% 6 " " +
- - "
2



" # \$
. 7 /
" -
, 3 /
\$77///2 2 7 87 7 /3 3 3 3 3
3 3# 3 3 3 39:3;<







!!

!!

"#!"



Http



Https



" # !" \$

" # ! "

7

= > ? = 8

@ " " A 9BB< + !
C

A -)

= > = 8 8

= "D2;

@) \$ 0

- 5" 2

, " E +

%) # #



"#!" % &

F 3 "

\$

-

"

G

, "

*

+

7

&

& \$ &8 H # & " =

% - #

F% F I



"#!"

'

A

"

"

\$, "

F

\$ (%

F

" - "

"

#

"

1

-

F%

%

% "-

" 1



"#!" (

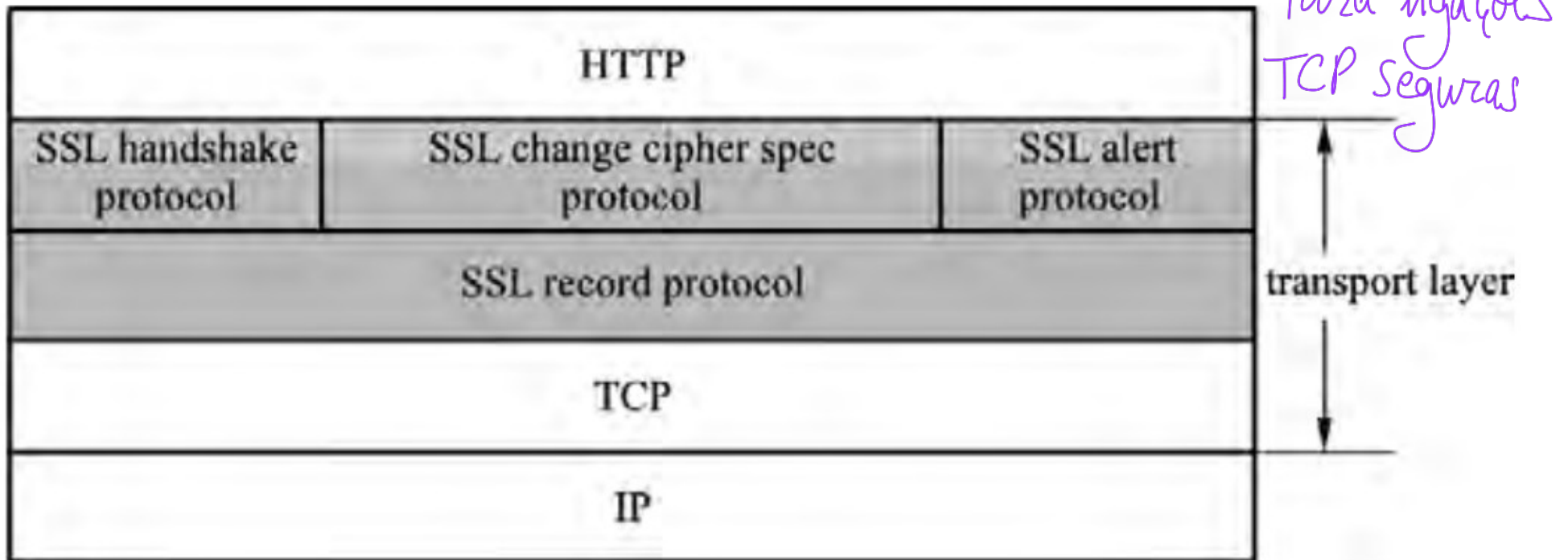
@ #
% "
. =

, + + =

Protocolos	Porto Normal	Porto TLS
HTTP	80	443
IMAP	143	993
POP3	110	995
SMTP	25	465
LDAP	389	636



"#!"





"#!"

Cipher Switch strings

, & ? 6 "\$

A

. - 6 \$

A "
% 7%
#

F %

= 92' TLS ECDHE ECDSA WITH AES 128 GCM SHA256

= 92D TLS AES 256 GCM SHA384

%

A

% " > % "-

% 1 " > % "-

A " -

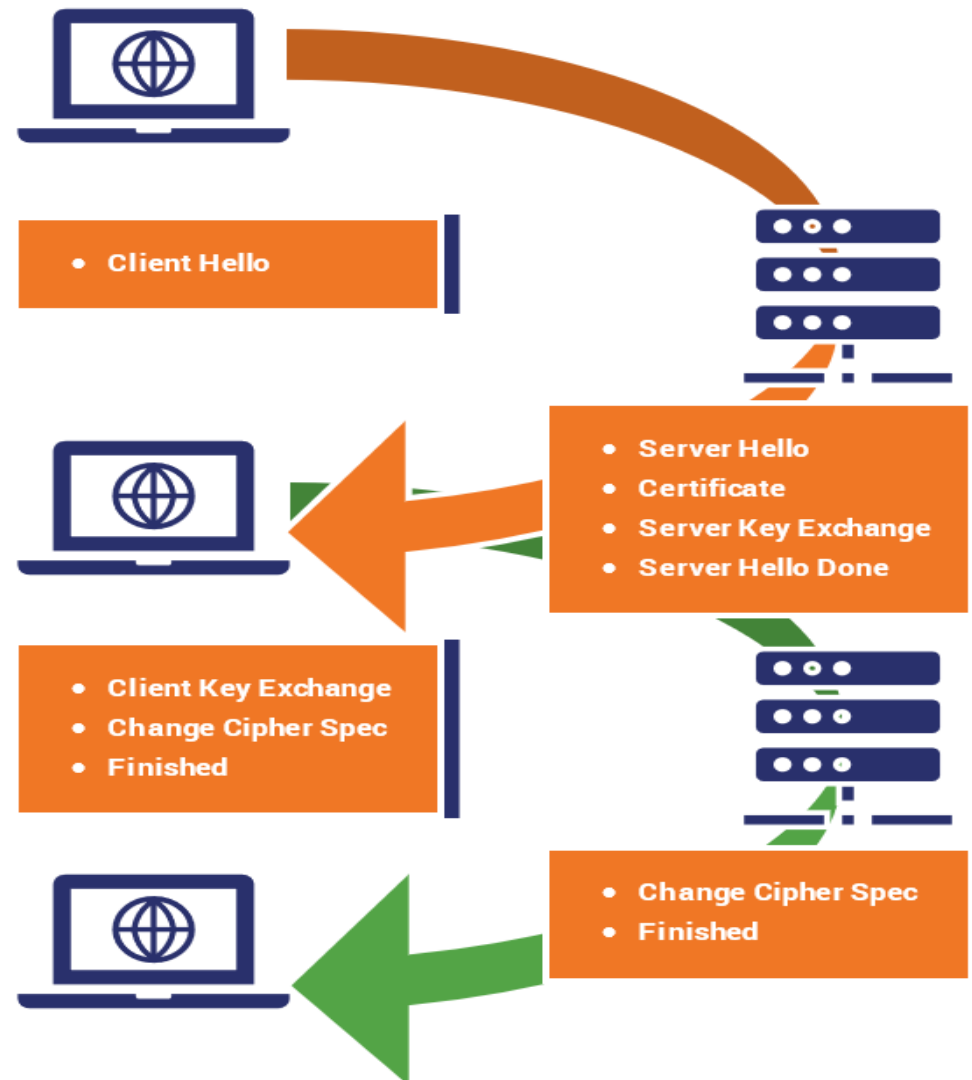
= 92'\$ @ \$ (% @&0

= 92D\$ " + 0 @&0

Para
MAC
↳ Para cifra/criptação



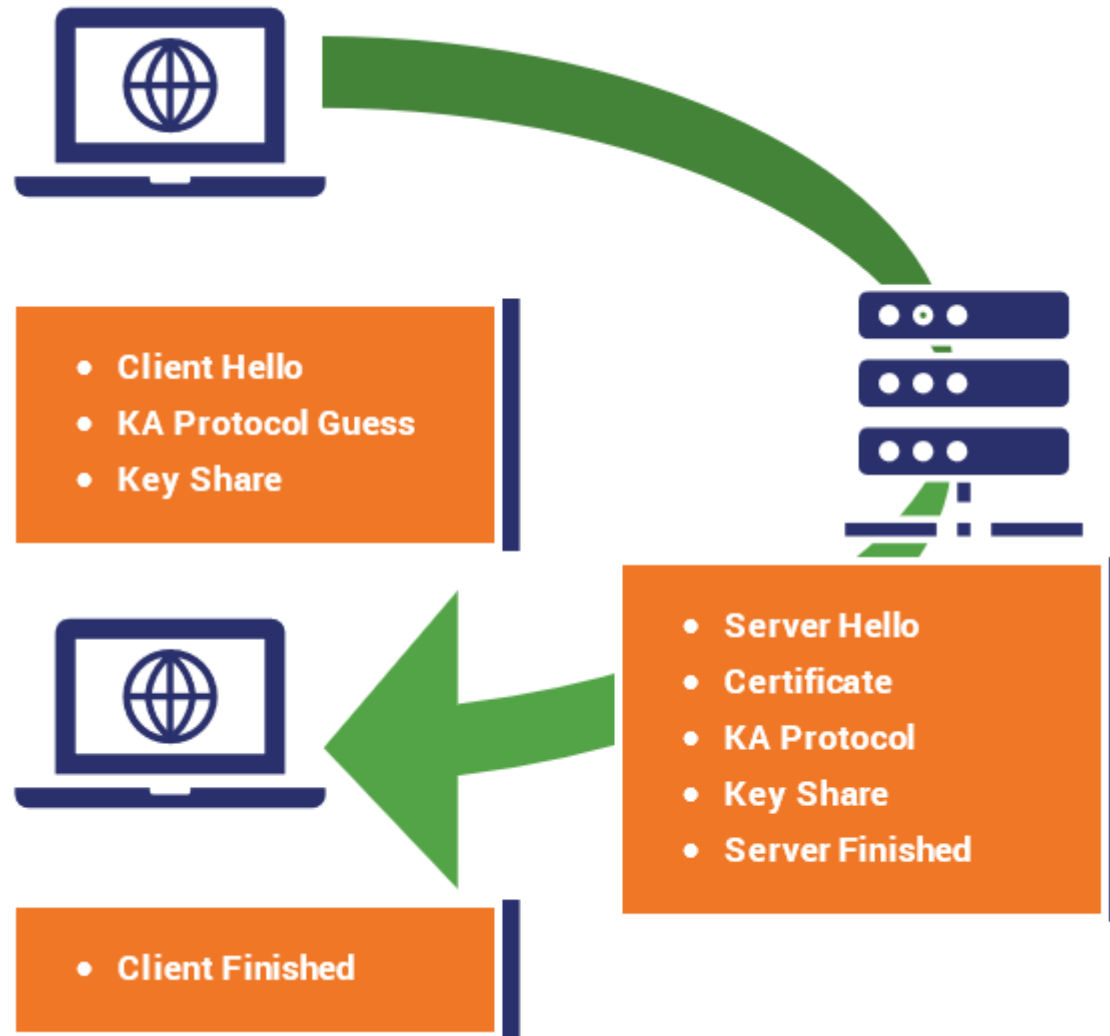
"#!"





"#!"

!")*+



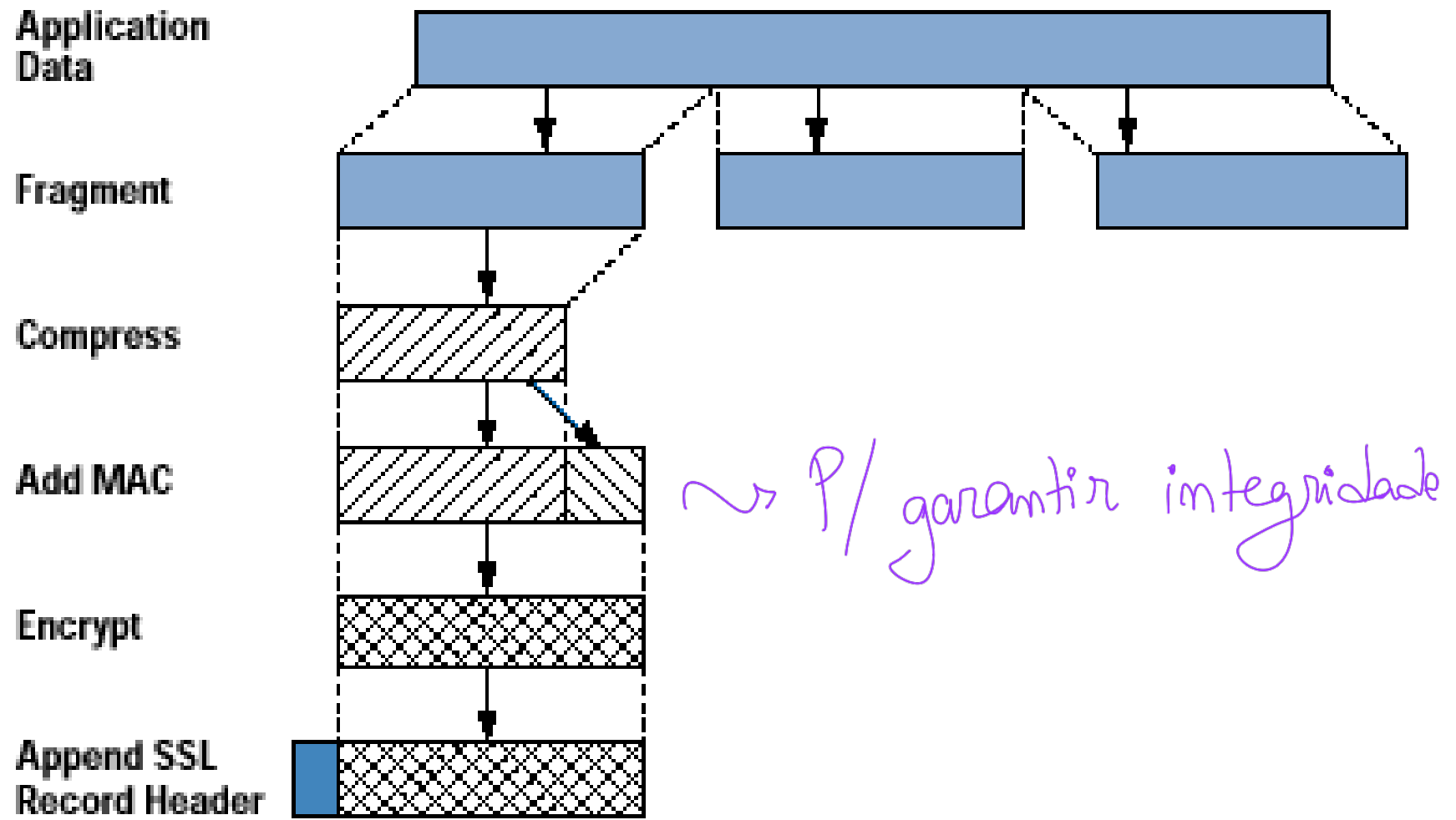


SSL/TLS: *Record Protocol*

*
2
\$
\$
" "
- "
\$ "
F % F% - 5



SSL/TLS: *Record Protocol*





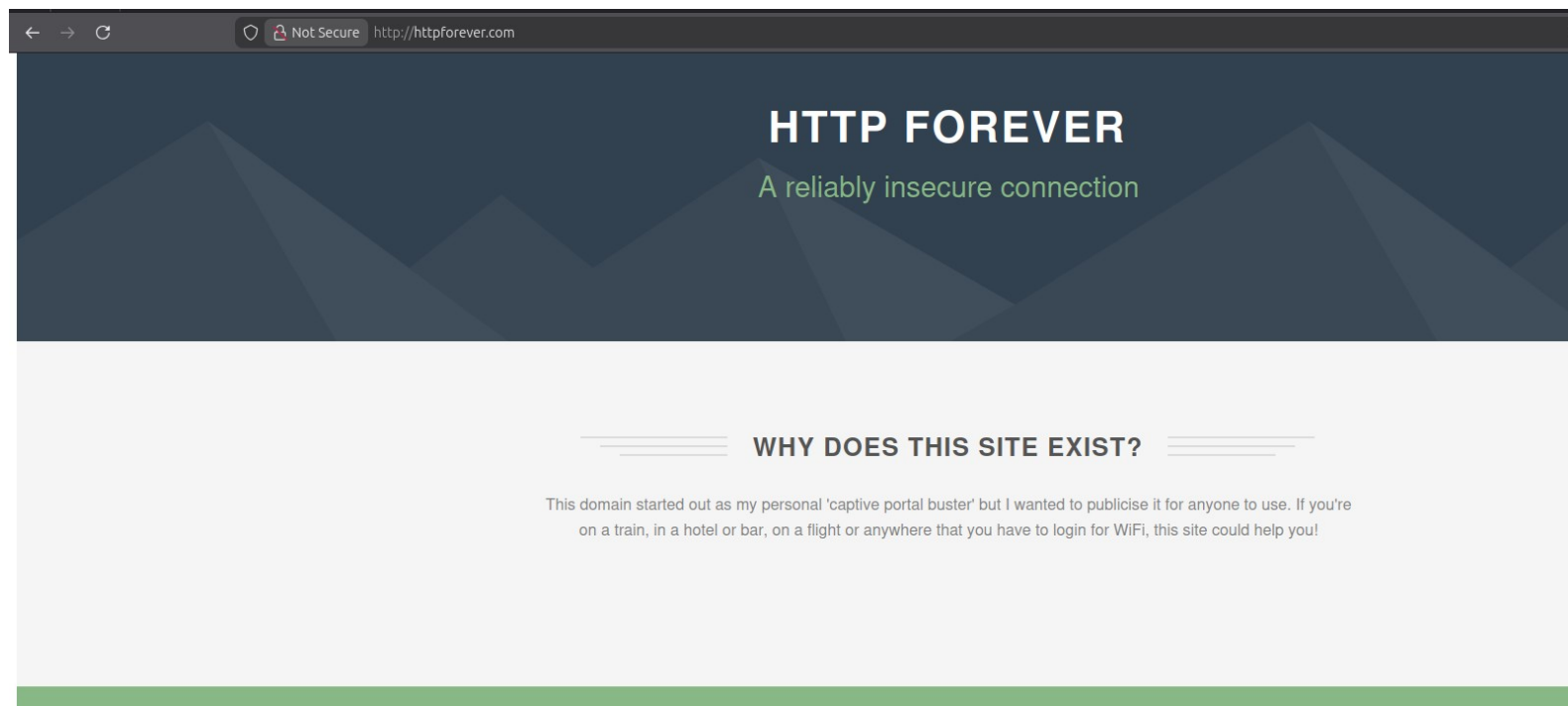
Navegação na Web: HTTP



http://info.cern.ch - home of the first website

From here you can:

- [Browse the first website](#)
- [Browse the first website using the line-mode browser simulator](#)
- [Learn about the birth of the web](#)
- [Learn about CERN, the physics laboratory where the web was born](#)





Navegação segura na Web: HTTPS





HTTPS: Exercício

C ? #

% J %"

//2 3 " 2

& 2

= K

G " = K

G) " K

" " / K



Certificado apresentado por cm-aveiro.pt

Certificate

*.cm-aveiro.pt		GoGetSSL RSA DV CA	USERTrust RSA Certification Authority
Subject Name			
Common Name	*.cm-aveiro.pt		
Issuer Name			
Country	LV		
Locality	Riga		
Organisation	GoGetSSL		
Common Name	GoGetSSL RSA DV CA		
Validity			
Not Before	Mon, 16 May 2022 00:00:00 GMT		
Not After	Mon, 22 May 2023 23:59:59 GMT		
Subject Alt Names			
DNS Name	*.cm-aveiro.pt		
DNS Name	cm-aveiro.pt		
Public Key Info			
Algorithm	RSA		
Key Size	4096		
Exponent	65537		
Modulus	EA:EA:EC:8D:37:F0:5D:77:DE:C9:64:FB:DC:BE:68:E3:F0:D9:92:AB:7E:EC:4B:9...		
Miscellaneous			
Serial Number	2C:AC:F3:C6:8A:9E:8F:06:AB:38:9F:AD:57:13:C1:8C		
Signature Algorithm	SHA-256 with RSA Encryption		
Version	3		
Download	PEM (cert) PEM (chain)		



Falha na autenticação do servidor

Warning: Potential Security Risk Ahead — Mozilla Firefox

Warning: Potential Security Certificate for portal.cm-viana-castelo.pt

Not Secure https://www.cm-viana-castelo.pt

Warning: Potential Security Risk Ahead

Firefox detected a potential security threat and did not continue to **www.cm-viana-castelo.pt**. If you visit this site, attackers could try to steal information like your passwords, emails, or credit card details.

[Learn more...](#)

[Go Back \(Recommended\)](#) [Advanced...](#)

www.cm-viana-castelo.pt uses an invalid security certificate.

The certificate is not trusted because it is self-signed.

Error code: [MOZILLA_PKIX_ERROR_SELF_SIGNED_CERT](#)

[View Certificate](#)

[Go Back \(Recommended\)](#) [Accept the Risk and Continue](#)



Porquê a falha na autenticação?

Certificate for portal.cm-viana-castelo.pt — Mozilla Firefox

Warning: Potential Security ... Certificate for portal.cm-viana- ...

Firefox about:certificate?cert=MIIDUzCCARYgAwIBAgICGjMwDQYJKoZIhvcNAQ...

portal.cm-viana-castelo.pt

Subject Name

Country	--
State/Province/County	SomeState
Locality	SomeCity
Organisation	SomeOrganization
Organisational Unit	SomeOrganizationalUnit
Common Name	portal.cm-viana-castelo.pt
Email Address	root@portal.cm-viana-castelo.pt

Issuer Name

Country	--
State/Province/County	SomeState
Locality	SomeCity
Organisation	SomeOrganization
Organisational Unit	SomeOrganizationalUnit
Common Name	portal.cm-viana-castelo.pt
Email Address	root@portal.cm-viana-castelo.pt

Validity

Not Before	Thu, 15 Nov 2012 12:29:18 GMT
Not After	Fri, 15 Nov 2013 12:29:18 GMT



Verificação de Sites

,

\$77///2 2 7

\$77/ ?2 7 7

\$77///2) 2 7

\$77///2 / 2 7

=

= H



VPN – Virtual Private Network (Rede Privada Virtual)



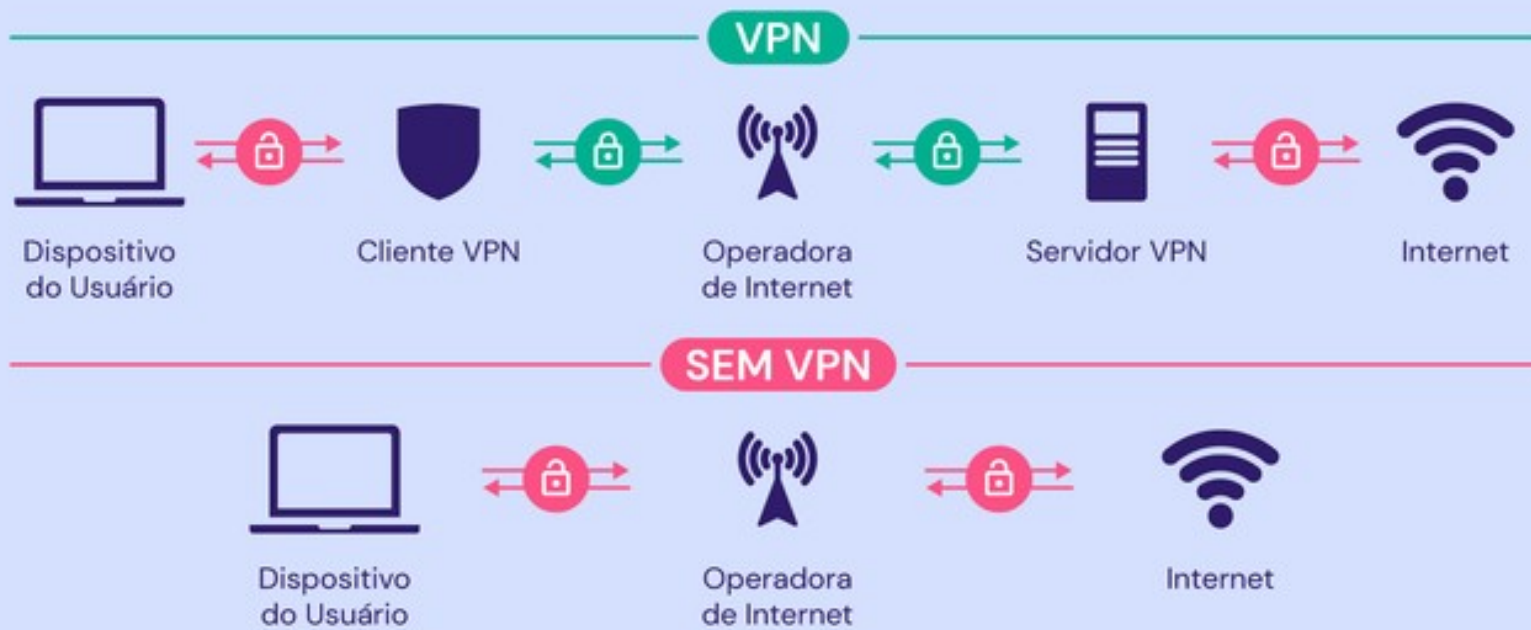


VPN: Acesso à Internet



Três. Dois. Online

Como Funciona uma VPN





Fornecedores de VPN para acesso à Internet





Rede Privada Virtual

(" "

1 # H

"_

) # E

L A J L A

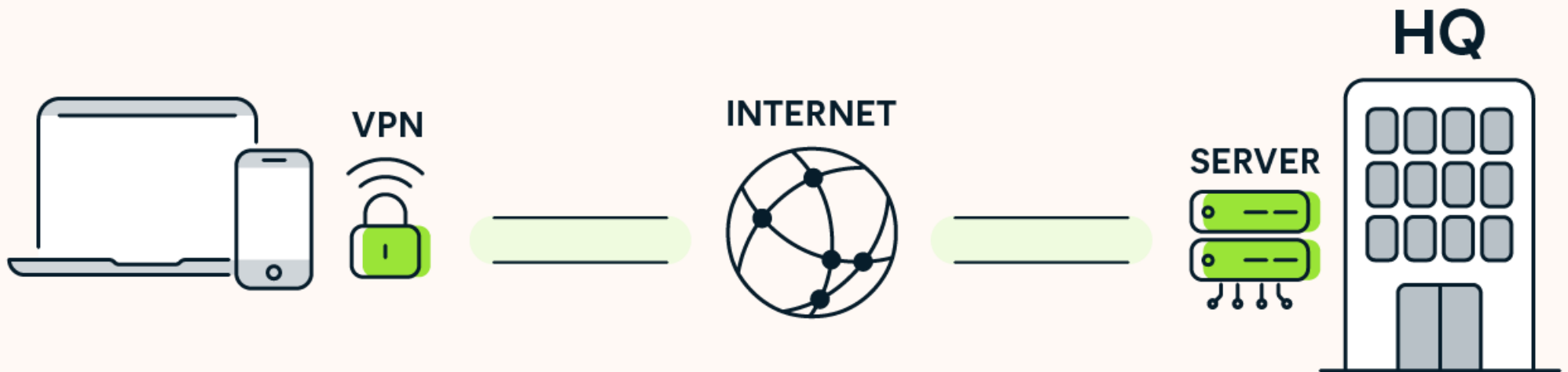
#

#

L A 3 3 L A



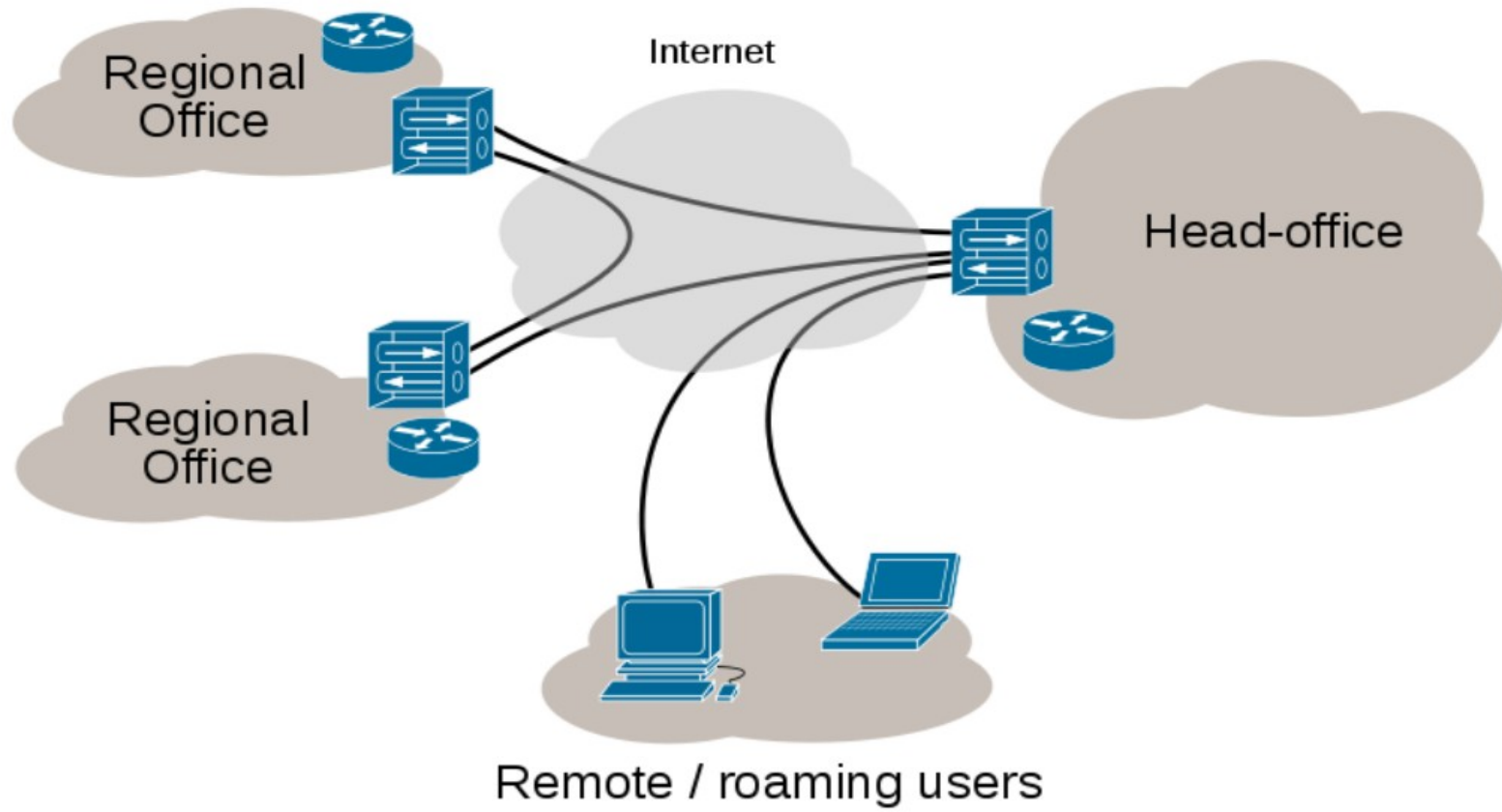
Interligação com utilizadores remotos





Tipos de VPNs

Internet VPN





Tecnologias

0
0 "
2
1
,
"
%
"
" " "
M / \$ % &% (%@ .
M " 1
F &
" 1
F #
#



Níveis de VPN e implementações

L A 5" ' '
=' ='

L A 5" D

L A =
, L A C

